POWERS OF AN ELEMENT OF THE JACOBSON RADICAL OF THE INTEGRAL CLOSURE OF A ONE-DIMENSIONAL SEMI-LOCAL DOMAIN LAND BACK IN THE DOMAIN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Structure of the integral closure of a one-dimensional Noetherian domain. Stacks Project, Algebra, Lemma algebra-lemma-integral-closure-Dedekind (tag 034W). Gives that B' is a semi-local Dedekind domain.
- Finiteness criteria for integral extensions and surjectivity of the spectrum map. Stacks Project, Algebra, Lemmas algebra-lemma-characterize-finite-in-terms-of-integral (tag 00GM) and algebra-lemma-integral-overring-surjective (tag 00GT). Make the intermediate ring A finite over B and show that A is semi-local.

Source reference. Original reference 09IG: sources/stacks-chow/tags.txt, source lines 12039–12039. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02JJ: sources/stacks-chow/tags.txt, source lines 3338–3338. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00GQ: sources/stacks-chow/tags.txt, source lines 616–616. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let B be a semi-local Noetherian domain of dimension 1. Let B' be the integral closure of B in its fraction field. Then B' is a semi-local Dedekind domain. Let x be a nonzero element of the Jacobson radical of B'. Then for every $y \in B'$ there exists an n such that $x^n y \in B$.*

Proof. Let $\mathfrak{m}_B$ be the Jacobson radical of B. The structure of B' results from Algebra, Lemma 09IG. Given $x, y \in B'$ as in the statement of the lemma consider the subring $B \subset A \subset B'$ generated by x and y. Then A is finite over B (Algebra, Lemma 02JJ). Since the fraction fields of B and A are the same we see that the finite module A/B is supported on the set of closed points of B. Thus $\mathfrak{m}_B^n A \subset B$ for a suitable n. Moreover, $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is surjective (Algebra, Lemma 00GQ), hence A is semi-local as well. It also follows that x is in the Jacobson radical $\mathfrak{m}_A$ of A. Note that $\mathfrak{m}_A = \sqrt{\mathfrak{m}_B A}$. Thus $x^m y \in \mathfrak{m}_B A$ for some m. Then $x^{nm} y \in B$. □

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